COSC 611 Big Data Analytics — Midterm Take-Home Assignment

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Overview

This report answers the four midterm questions. Every concept is illustrated with (a) a **real, published dataset**, (b) a hand computation that can be checked with a calculator, and (c) the same computation carried out on **Apache Spark** (PySpark, `pyspark.ml` / Spark MLlib). The Spark programs are submitted alongside this report:

Question	Spark program	Captured output	Real datasets used
Q1 Similarity measures	`code/q1_similarity.py`	`outputs/q1_similarity.txt`	MovieLens ratings (GroupLens)
Q2 Hypothesis testing	`code/q2_hypothesis_tests.py`	`outputs/q2_hypothesis_tests.txt`	UCBAdmissions, sleep, ToothGrowth, PlantGrowth
Q3 Clustering	`code/q3_clustering.py`	`outputs/q3_clustering.txt`	Iris, Old Faithful
Q4 Classification	`code/q4_classification.py`	`outputs/q4_classification.txt`	Play Tennis, Iris, Spambase (spam7)

The data files are in `data/`, and their origins are listed in `data/SOURCES.md` and in the References.

Environment used to produce the outputs: Apache Spark 4.0.1 (PySpark) in `local[*]` mode, Python 3.13, OpenJDK 21, SciPy 1.18 (used only to cross-check p-values). Run each script from the `cosc611-midterm` folder with `spark-submit code/qN_*.py` or `python3 code/qN_*.py`. All numbers quoted below are copied from the captured outputs.

Question 1 — Similarity Measures

1.1 What is a similarity measure?

A **similarity measure** is a function $s(x, y)$ that takes two objects (two customers, two documents, two images, two gene sequences, …) and returns a number that says *how alike they are*. The higher the value, the more alike the objects are. Its opposite is a **distance (dissimilarity) measure** $d(x, y)$, where a *smaller* value means more alike. One can usually be turned into the other, for example $s = \frac{1}{1+d}$ or $s = 1 - d$ when $d \in [0, 1]$.

A good similarity measure usually has these properties:

- **Bounded/normalised**, often in $[0, 1]$ or $[-1, 1]$.
- **Symmetric**: $s(x, y) = s(y, x)$.
- **Maximal for identical objects**: $s(x, x)$ is the largest possible value.

A *distance metric* must also satisfy non-negativity, $d(x, y) = 0 \Leftrightarrow x = y$, and the triangle inequality $d(x, z) \leq d(x, y) + d(y, z)$.

1.2 Why are similarity measures necessary?

Many Big Data analytics tasks come down to the question *"which items are close to which?"*:

- **Recommender systems.** Collaborative filtering finds users whose rating vectors are similar to yours and recommends what they liked ("people like you also bought …").
- **Clustering.** K-Means, hierarchical clustering and DBSCAN group points that are close to each other (Question 3), so they need a distance.
- **Classification.** k-Nearest-Neighbours labels a point by the labels of its most similar neighbours. SVM kernels are similarity functions.
- **Information retrieval and search.** Search engines rank documents by their cosine similarity to the query vector (TF-IDF).
- **De-duplication and plagiarism detection.** Near-duplicate web pages and records are found with Jaccard similarity and MinHash.
- **Anomaly detection.** A point that is dissimilar to everything else is an outlier.

The right measure depends on the data type: numeric vectors, sets, text, or binary attributes. A wrong choice gives misleading neighbours, as the example below shows. At Big Data scale, comparing all n^2 pairs is too expensive, so Spark provides **Locality-Sensitive Hashing (LSH)**. LSH finds similar pairs approximately without comparing every pair.

1.3 Example data

The data is the **MovieLens** "latest-small" ratings dataset collected by GroupLens at the University of Minnesota (Harper & Konstan, 2015), in the copy distributed with the R package `dslabs`. It contains **100,004 ratings** (0.5–5 stars) by **671 users** on **9,066 movies**.

For the hand computations, the program takes the **6 most-rated movies** and the **first 5 users (by user ID) who rated at least 4 of them**. A 0 means the user did not rate the movie.

User	Forrest Gump	Pulp Fiction	Shawshank Redemption	Silence of the Lambs	Star Wars IV	Jurassic Park
user2	3.0	4.0	0	3.0	0	4.0
user3	5.0	4.5	5.0	3.0	0	0
user4	5.0	5.0	0	0	5.0	5.0
user7	3.0	0	5.0	0	5.0	4.0
user8	4.0	4.0	5.0	4.5	3.5	0

These movies were rated by 341, 324, 311, 304, 291 and 274 users respectively. In the hand computations below, x = user3 = (5, 4.5, 5, 3, 0, 0) and y = user8 = (4, 4, 5, 4.5, 3.5, 0).

Filling unrated movies with 0 is a common simplification. It treats “not rated” as “disliked”, which real recommender systems avoid by comparing only co-rated items or by mean-centring.

1.4 Types of similarity measures, with computations

(a) Euclidean distance (L2)

The straight-line distance between two points:

$$d_E(x,y) = \sqrt{\sum_i (x_i - y_i)^2}, \qquad s_E = \frac{1}{1 + d_E}$$

$x - y = (1, 0.5, 0, -1.5, -3.5, 0)$, so $d_E = \sqrt{1 + 0.25 + 0 + 2.25 + 12.25 + 0} = \sqrt{15.75} = 3.9686$ and $s_E = 1/(1 + 3.9686) = 0.2013$.

Euclidean distance is sensitive to scale and magnitude. It is a natural fit for dense, low-dimensional numeric data such as K-Means features.

(b) Manhattan distance (L1, city-block)

$$d_M(x,y) = \sum_i |x_i - y_i|$$

$d_M = 1 + 0.5 + 0 + 1.5 + 3.5 + 0 = 6.5$. Manhattan distance is less affected by one large difference than Euclidean distance, so it is more robust to outliers.

(c) Cosine similarity

The cosine of the angle between two vectors. It measures orientation rather than magnitude:

$$\cos(x,y) = \frac{x \cdot y}{\|x\| \, \|y\|}$$

- $x \cdot y = 20 + 18 + 25 + 13.5 + 0 + 0 = 76.5$
- $\|x\| = \sqrt{25 + 20.25 + 25 + 9} = \sqrt{79.25} = 8.9022$
- $\|y\| = \sqrt{16 + 16 + 25 + 20.25 + 12.25} = \sqrt{89.5} = 9.4604$
- $\cos(x,y) = 76.5/(8.9022 \times 9.4604) = 76.5/84.219 = 0.9083$

The range is $[-1, 1]$, or $[0, 1]$ for non-negative data. Cosine is the standard measure for text (TF-IDF vectors) and for sparse, high-dimensional data such as a user's ratings over 9,066 movies.

(d) Pearson correlation coefficient

Cosine similarity applied after **mean-centring** each vector. This removes each user's personal rating bias (a "harsh" rater versus a "generous" one):

$$r(x, y) = \frac{\sum_i (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_i (x_i - \bar{x})^2}\sqrt{\sum_i (y_i - \bar{y})^2}}$$

- $\bar{x} = 17.5/6 = 2.9167$ and $\bar{y} = 21/6 = 3.5$
- Deviations: $x' = (2.083, 1.583, 2.083, 0.083, -2.917, -2.917)$ and $y' = (0.5, 0.5, 1.5, 1.0, 0, -3.5)$
- $\sum x'y' = 1.042 + 0.792 + 3.125 + 0.083 + 0 + 10.208 = 15.25$
- $\sum x'^2 = 28.208$ and $\sum y'^2 = 16$
- $r = 15.25/\sqrt{28.208 \times 16} = 15.25/21.245 = 0.7178$

The range is $[-1, 1]$, where -1 means opposite tastes.

(e) Jaccard similarity (sets)

Jaccard similarity is used when the data is a **set** or binary presence/absence (which movies were rated, which genres a movie has, which words appear):

$$J(S, T) = \frac{|S \cap T|}{|S \cup T|}$$

- user3 rated {Forrest Gump, Pulp Fiction, Shawshank, Silence of the Lambs}.
- user8 rated those four plus Star Wars IV.
- Intersection = 4 and union = 5, so $J = 4/5 = 0.8$.

Genre sets (real MovieLens genre labels):

- The Matrix = {Action, Sci-Fi, Thriller}
- Jurassic Park = {Action, Adventure, Sci-Fi, Thriller}
- So $J = 3/4 = 0.75$.

Other measures for binary data include the **Simple Matching Coefficient**, which also counts shared 0s, the **Hamming distance**, which counts positions that differ, and the **Dice coefficient** $2|S \cap T|/(|S| + |T|)$. For text there is also **edit (Levenshtein) distance**.

1.5 Computation on Spark

`code/q1_similarity.py` reads the 100,004 ratings with `spark.read.csv` and selects the 5×6 matrix with DataFrame operations (`groupBy`, `orderBy`, `isin`). It then **self cross-joins** the users to form every pair and evaluates each measure on the pair. This is the distributed form of the n^2 comparison.

```
        a, b = df.alias("a"), df.alias("b")
        pairs = a.crossJoin(b).where(F.col("a.user") < F.col("b.user"))
        result = (pairs.withColumn("euclidean", euclidean("x", "y"))
                       .withColumn("cosine", cosine("x", "y"))
```

```
                    .withColumn("pearson", pearson("x", "y"))
                    .withColumn("jaccard", jaccard("x", "y")) ...)
```

Spark output (all 10 pairs):

u1	u2	Euclidean	1/(1+d)	Manhattan	Cosine	Pearson	Jaccard
user2	user3	6.7268	0.1294	11.5	0.6672	0.0528	0.6000
user2	user4	6.3246	0.1365	12.0	0.7778	0.3467	0.6000
user2	user7	8.6603	0.1035	17.0	0.4082	−0.6801	0.3333
user2	user8	7.5166	0.1174	15.0	0.6204	**−0.4504**	0.5000
user3	user4	9.1788	0.0982	18.5	0.5336	−0.3533	0.3333
user3	user7	8.6168	0.1040	18.5	0.5188	−0.3483	0.3333
user3	user8	**3.9686**	**0.2013**	**6.5**	**0.9083**	**0.7178**	**0.8000**
user4	user7	7.4162	0.1188	13.0	0.6928	0.1115	0.6000
user4	user8	8.6313	0.1038	18.0	0.6078	−0.5413	0.5000
user7	user8	7.4498	0.1183	15.0	0.6652	−0.2413	0.5000

The Spark values match the hand computations for user3–user8 exactly.

The script also shows three **Spark-native** ways of computing these measures:

1. **Cosine via `Normalizer`.** `Normalizer(p=2.0)` scales every vector to unit length, so a plain dot product of two unit vectors *is* the cosine. Spark gives the same values, for example user3–user8 = 0.9083.

2. **Pearson via `pyspark.ml.stat.Correlation.corr`.** This computes the full correlation matrix in one distributed pass. (Spark correlates columns, so the matrix is transposed so that each user is a column.)

   ```
            user2   user3   user4   user7   user8
   user2    1.000   0.053   0.347  -0.680  -0.450
   user3    0.053   1.000  -0.353  -0.348   0.718
   user4    0.347  -0.353   1.000   0.111  -0.541
   user7   -0.680  -0.348   0.111   1.000  -0.241
   user8   -0.450   0.718  -0.541  -0.241   1.000
   ```

3. **Cosine nearest neighbours at full scale.** Each of the 671 users becomes a sparse vector over all movies, built with `groupBy` + `collect_list` and a `SparseVector`. After normalising, user8's vector is broadcast and dotted with every other user. The 5 most similar users to user8 out of all 670 are users 520 (0.4403), 216 (0.4327), 562 (0.4294), 379 (0.4217) and 282 (0.4067). These scores are lower than in the 5-movie example because two users share only a small fraction of the 9,066 movies.

Jaccard at scale: MinHash LSH on genre sets. The script turns the genre list of each of the 9,049 movies that have genres into a binary vector over the 19 genres (`CountVectorizer(binary=True)`). The exact Jaccard similarities for five well-known movies are:

Movie 1	Movie 2	Jaccard
Jurassic Park	The Matrix	0.7500

Movie 1	Movie 2	Jaccard
Forrest Gump	Pulp Fiction	0.3333
The Matrix	Pulp Fiction	0.1667
Jurassic Park	Pulp Fiction	0.1429
Toy Story	Forrest Gump / Jurassic Park / Pulp Fiction	0.1250
Forrest Gump	The Matrix / Jurassic Park	0.0

`MinHashLSH` hashes each set so that the probability that two sets get the same hash *equals* their Jaccard similarity. Only pairs that collide in some hash table are compared:

- `approxNearestNeighbors` for *The Matrix* returns movies with exactly the same genres (Jaccard distance 0), such as *The Terminator*, *Blade Runner* and *Equilibrium*.
- `approxSimilarityJoin` with distance ≤ 0.2 (Jaccard ≥ 0.8) found **1,413,821** pairs without comparing all 40,937,676 pairs one by one.
- An exact brute-force check, also done in Spark, gives the same **1,413,821**, so the LSH recall is 1.0 here. Most of these pairs (1,406,827) are movies with identical genre sets, for example two movies labelled only “Drama”.

1.6 Interpretation: the choice of measure matters

- For user8, every measure picks **user3** as the most similar user: the same movies, rated in a similar pattern.
- **user2 and user8** show why the measure matters. Their cosine similarity is fairly high (0.62), but their Pearson correlation is **negative (−0.45)**. Ratings are all non-negative, so the cosine of raw ratings is always ≥ 0. It mostly reflects that both users rate movies highly in general. Once each user’s average is subtracted, Pearson shows that the two users *prefer different movies*.
- **Jaccard** ignores the ratings entirely and only uses *which* movies were rated. That makes it the right choice for genre sets, but it would call a lover and a hater of the same movies “identical”.

In general:

- Use **cosine** for text and sparse vectors.
- Use **Pearson** for ratings with user bias.
- Use **Euclidean or Manhattan** for dense numeric features, after scaling them.
- Use **Jaccard** for sets or binary data.

Question 2 — Hypothesis Testing

2.1 What is hypothesis testing?

Hypothesis testing is a formal procedure for using **sample data** to decide whether a claim about a **population** is supported. Analysts usually only see a sample, so the question is whether an observed effect is real or could just be random sampling noise. The steps are:

1. **State the hypotheses.** The *null hypothesis* H_0 is the "no effect / no difference / independent" claim. The *alternative hypothesis* H_1 is what we suspect is true.
2. **Choose the significance level** α, usually 0.05. This is the probability of rejecting H_0 when it is actually true, called a **Type I error** (false positive). Failing to reject a false H_0 is a **Type II error** (false negative). The power of a test is $1 - P$(Type II).
3. **Choose the test and compute the test statistic** (χ^2, t or F) from the sample.
4. **Find the p-value**: the probability, *assuming* H_0 *is true*, of seeing a statistic at least as extreme as the one observed. Equivalently, compare the statistic with a **critical value** from the statistic's distribution.
5. **Decide.** If $p < \alpha$, reject H_0 (the result is statistically significant). Otherwise, *fail to reject* H_0. Failing to reject does not prove H_0 true.

Tests can be **one-tailed** or **two-tailed**, depending on whether H_1 specifies a direction.

2.2 Distinguishing the χ^2, t and F tests

	Chi-square (χ^2) test	Student's t-test	F-test
Question answered	Are two *categorical* variables independent? Does an observed frequency distribution fit an expected one?	Is a *mean* different from a value, or are two *means* different?	Are two *variances* equal? Are 3 or more means equal (ANOVA)?
Data type	Counts/frequencies of categories	Continuous, approximately normal	Continuous, approximately normal
Statistic	$\chi^2 = \sum \frac{(O-E)^2}{E}$	$t = \frac{\bar{x}_1 - \bar{x}_2}{s_p\sqrt{1/n_1 + 1/n_2}}$	$F = \frac{s_1^2}{s_2^2}$, or $F = \frac{MS_{between}}{MS_{within}}$
Distribution	χ^2 with $(r-1)(c-1)$ df	t with $n_1 + n_2 - 2$ df (two-sample) or $n - 1$ df (one-sample)	F with (d_1, d_2) df
Shape	Right-skewed, values ≥ 0	Symmetric bell, heavier tails than the normal	Right-skewed, values ≥ 0
Parametric?	Non-parametric (no normality assumption)	Parametric	Parametric
Typical use	Survey answers vs demographic; click vs device	A/B test of average spend; before/after	Comparing variability of processes; ANOVA across many groups; overall significance of a regression
Spark support	`pyspark.ml.stat.ChiSquareTest` (built in)	No built-in test. Computed from distributed aggregates (mean, var, count)	`ANOVATest` and `FValueTest` (`org.apache.spark.ml.stat`), or from aggregates

How they relate. t^2 with v df equals F with $(1, v)$ df, so a two-group ANOVA is the same test as a two-sample t-test. The χ^2 distribution is the distribution of a sum of squared standard normals, and F is a ratio of two scaled χ^2 variables.

2.3 Chi-square test of independence — UC Berkeley admissions

Data. `UCBAdmissions` records the 4,526 applicants to the six largest graduate departments at UC Berkeley in Fall 1973, by gender and admission decision (Bickel, Hammel & O'Connell, 1975). Summed over the departments:

Observed	Admitted	Rejected	Row total
Male	1198	1493	2691
Female	557	1278	1835
Column total	1755	2771	N = 4526

H_0 : admission is independent of gender. H_1 : they are associated.

Hand computation. The expected count for each cell is $E = \frac{\text{row total} \times \text{column total}}{N}$:

Cell	O	E	$(O-E)^2/E$
Male, Admitted	1198	2691·1755/4526 = 1043.461	22.8875
Male, Rejected	1493	2691·2771/4526 = 1647.539	14.4957
Female, Admitted	557	1835·1755/4526 = 711.539	33.5642
Female, Rejected	1278	1835·2771/4526 = 1123.461	21.2578

$$\chi^2 = 22.8875 + 14.4957 + 33.5642 + 21.2578 = 92.21$$

The degrees of freedom are $(2-1)(2-1) = 1$, and the critical value is $\chi^2_{0.05,\,1} = 3.8415$. Since $92.21 \gg 3.84$, **reject** H_0 : overall, 44.5% of men but only 30.4% of women were admitted.

Spark. `groupBy().pivot()` builds the table, and a groupBy/join pipeline reproduces the $(O-E)^2/E$ column above. For MLlib's built-in test, each frequency row is expanded into one row per applicant with `explode(sequence(1, Freq))`, which gives 4,526 rows:

```
r = ChiSquareTest.test(vec, "features", "label").head()
# Spark ChiSquareTest: statistic = 92.2053, df = 1, p-value = 0.000e+00
# scipy check:          statistic = 92.2053, p-value = 7.814e-22
```

Spark prints the p-value as 0 because the true value, 7.8×10^{-22}, is below the precision of its χ^2 tail computation. SciPy gives the exact figure.

Simpson's paradox. Running the same Spark `ChiSquareTest` *inside each department* reverses the story:

Dept	Male admit rate	Female admit rate	χ^2	p-value	Decision
A	0.621	0.824	17.248	<0.0001	Reject H_0 (favours women)
B	0.630	0.680	0.254	0.6145	Fail to reject
C	0.369	0.341	0.754	0.3854	Fail to reject
D	0.331	0.349	0.298	0.5852	Fail to reject

Dept	Male admit rate	Female admit rate	χ^2	p-value	Decision
E	0.277	0.239	1.001	0.3171	Fail to reject
F	0.059	0.070	0.384	0.5354	Fail to reject

Women mostly applied to departments with low admission rates (C–F). The overall association is real, but it is explained by *department*, a confounding variable, not by bias within departments. This was the conclusion of Bickel et al. (1975). A significant χ^2 shows association, not cause.

2.4 Student's t-test — the sleep data

Data. `sleep` is the dataset that W. S. Gosset ("Student") used in the 1908 paper that introduced the t-test. It is taken from Cushny & Peebles (1905). Ten patients each received two sleep-inducing drugs, and the extra hours of sleep compared with a control were recorded:

Patient	1	2	3	4	5	6	7	8	9	10
Drug 1	0.7	−1.6	−0.2	−1.2	−0.1	3.4	3.7	0.8	0.0	2.0
Drug 2	1.9	0.8	1.1	0.1	−0.1	4.4	5.5	1.6	4.6	3.4
Difference	−1.2	−2.4	−1.3	−1.3	0.0	−1.0	−1.8	−0.8	−4.6	−1.4

$H_0 : \mu_1 = \mu_2$. $H_1 : \mu_1 \neq \mu_2$ (two-tailed).

(a) Independent two-sample t-test (pooled variance).

- $\bar{x}_1 = 0.75$, $s_1^2 = 3.2006$; $\bar{x}_2 = 2.33$, $s_2^2 = 4.0090$
- Pooled variance: $s_p^2 = \frac{9(3.2006) + 9(4.0090)}{18} = 3.6048$
- Standard error: $SE = \sqrt{3.6048(1/10 + 1/10)} = 0.8491$
- $t = (0.75 - 2.33)/0.8491 = -1.8608$, with $df = 18$

The critical value is $t_{0.025,\,18} = 2.1009$ and $p = 0.0792$. Since $|t| < 2.1009$, we **fail to reject** H_0 .

(b) Paired t-test. The *same* ten patients took both drugs, so the correct analysis uses the ten differences d_i :

- $\bar{d} = -1.58$ and $s_d = 1.2300$
- $t = \bar{d}/(s_d/\sqrt{n}) = -1.58/(1.23/\sqrt{10}) = -4.0621$, with $df = 9$

The critical value is $t_{0.025,\,9} = 2.2622$ and $p = 0.0028$, so we **reject** H_0 : drug 2 gives more extra sleep.

The two versions reach opposite conclusions on the same data. The paired test removes the large patient-to-patient variability, which the independent test wrongly treats as noise. Choosing the test that matches the study design is part of hypothesis testing.

Spark. MLlib has no t-test class, but the test only needs counts, means and sample variances. These come from **one distributed aggregation**: `groupBy("group").agg(count, mean, var_samp)` for (a), and `pivot` + `mean`/`stddev_samp` of the differences for (b). Only the final scalar formula and the t-distribution tail probability run on the driver:

```
(a) Spark-computed t = -1.8608, df = 18, p = 0.07919   | scipy: t = -1.8608, p = 0.07919
```

```
(b) Spark-computed t = -4.0621, df = 9,  p = 0.002833  | scipy: t = -4.0621, p = 0.002833
```

2.5 F-test

(a) F-test for equality of two variances — ToothGrowth

Data. `ToothGrowth` (Crampton, 1947) gives the length of odontoblasts (the cells responsible for tooth growth) in 60 guinea pigs. Each received vitamin C either as orange juice (OJ, n = 30) or as ascorbic acid (VC, n = 30).

$H_0 : \sigma^2_{OJ} = \sigma^2_{VC}$. $H_1 : \sigma^2_{OJ} \neq \sigma^2_{VC}$.

- Spark aggregates give $s^2_{OJ} = 43.633$ and $s^2_{VC} = 68.327$.
- $F = 43.633/68.327 = 0.6386$, with $df = (29, 29)$.
- The two-tailed critical values are $F_{0.025;\,29,\,29} = 0.4760$ and $F_{0.975;\,29,\,29} = 2.1010$.
- $p = 0.2331$.

Since $0.476 < 0.6386 < 2.101$, we **fail to reject** H_0: there is no evidence that the two supplement types give different variability. Equal variances are one of the assumptions of the pooled t-test and of ANOVA, so this F-test is often run first as a check.

(b) One-way ANOVA F-test — PlantGrowth

Data. `PlantGrowth` (Dobson, 1983) records the dried weight of 30 plants: 10 controls and 10 under each of two treatments.

Group	Weights	Mean
ctrl	4.17, 5.58, 5.18, 6.11, 4.50, 4.61, 5.17, 4.53, 5.33, 5.14	5.032
trt1	4.81, 4.17, 4.41, 3.59, 5.87, 3.83, 6.03, 4.89, 4.32, 4.69	4.661
trt2	6.31, 5.12, 5.54, 5.50, 5.37, 5.29, 4.92, 6.15, 5.80, 5.26	5.526

$H_0 : \mu_{ctrl} = \mu_{trt1} = \mu_{trt2}$. H_1: at least one mean differs.

Hand computation. The grand mean is 5.073.

- $SSB = 10[(5.032 - 5.073)^2 + (4.661 - 5.073)^2 + (5.526 - 5.073)^2] = 10(0.00168 + 0.16974 + 0.20521) = 3.7663$
- $SSW = \sum_g \sum_i (x_{gi} - \bar{x}_g)^2 = 9(0.33999 + 0.62992 + 0.19587) = 10.4921$
- $MSB = 3.7663/2 = 1.8832$ and $MSW = 10.4921/27 = 0.3886$
- $F = 1.8832/0.3886 = 4.846$, with $df = (2, 27)$

The critical value is $F_{0.05;\,2,\,27} = 3.354$ and $p = 0.0159$, so **reject** H_0: the treatments affect plant weight. Treatment 2 has the highest mean and treatment 1 the lowest.

Spark. The ANOVA table is computed with DataFrame aggregations, and Spark's `org.apache.spark.ml.stat.ANOVATest` is also called. That class has no Python wrapper in PySpark 4.0, so the script calls it through the JVM gateway:

```
        jdf = spark._jvm.org.apache.spark.ml.stat.ANOVATest.test(anova_df._jdf, "features",
"label")
        # Spark ANOVATest: F = 4.8461, df = (2, 27), p-value = 0.01591
        # scipy check:     F = 4.8461, p-value = 0.01591
```

2.6 Summary of results

Test	Dataset	Statistic	df	p-value	Critical value (α=0.05)	Decision
χ^2 independence	UCBAdmissions	92.21	1	7.8×10^{-22}	3.84	Reject H_0
Two-sample t (pooled)	sleep	−1.86	18	0.0792	±2.10	Fail to reject H_0
Paired t	sleep	−4.06	9	0.0028	±2.26	Reject H_0
F (variances)	ToothGrowth	0.639	(29, 29)	0.2331	0.476 / 2.101	Fail to reject H_0
ANOVA F	PlantGrowth	4.846	(2, 27)	0.0159	3.354	Reject H_0

Question 3 — Supervised vs Unsupervised Learning, and Clustering

3.1 Supervised versus unsupervised learning

	Supervised learning	Unsupervised learning
Training data	**Labelled**: each example x comes with the correct answer y	**Unlabelled**: only x is given
Goal	Learn a mapping $f: x \rightarrow y$ that predicts labels or values for new data	Discover hidden *structure*: groups, density, low-dimensional representation, associations
Feedback	An error signal against known labels (loss)	No "right answer". Quality is judged by internal criteria (cohesion and separation) or by a domain expert
Typical tasks	Classification (spam/not spam), regression (house price)	Clustering (customer segments), dimensionality reduction (PCA), association rules (market basket), anomaly detection
Example algorithms	Naïve Bayes, SVM, Random Forest, logistic and linear regression, neural networks	K-Means, Bisecting K-Means / hierarchical, Gaussian Mixture, DBSCAN, PCA, FP-Growth
Evaluation	Accuracy, precision/recall, F1, AUC, RMSE on a held-out test set	Within-cluster SSE, silhouette score, log-likelihood, visual inspection
Cost of data	Labels are often expensive (human annotation)	Raw data is cheap and plentiful, which suits Big Data
Spark MLlib	`pyspark.ml.classification`, `pyspark.ml.regression`	`pyspark.ml.clustering`, `pyspark.ml.feature.PCA`, `pyspark.ml.fpm`

An analogy: supervised learning is like studying with an answer key. Unsupervised learning is like being handed a pile of documents and asked to sort them into sensible piles without being told what the piles should be. Semi-supervised and reinforcement learning sit between these two.

3.2 Example data

Iris (Anderson, 1935; Fisher, 1936) contains 150 iris flowers, 50 from each of three species: *setosa*, *versicolor* and *virginica*. Each flower has four measurements in cm: sepal length, sepal width, petal length and petal width. The species means are:

Species	Sepal L	Sepal W	Petal L	Petal W
setosa	5.006	3.428	1.462	0.246
versicolor	5.936	2.770	4.260	1.326
virginica	6.588	2.974	5.552	2.026

The algorithms see **only the four measurements**. The species label is held back and used only afterwards, to check how well each algorithm discovered the true groups. This is a standard way to evaluate an unsupervised method on data where the answer happens to be known.

Old Faithful (R `faithful`; Härdle, 1991; see also Azzalini & Bowman, 1990) records 272 eruptions of the Old Faithful geyser in Yellowstone National Park: eruption duration and waiting time, in minutes. It is used below to show soft clustering with a Gaussian Mixture.

All algorithms run in `code/q3_clustering.py`, with `VectorAssembler` building the feature vector and `ClusteringEvaluator` (silhouette) used for evaluation.

3.3 Algorithm 1 — K-Means (partitional, hard clustering)

Idea. K-Means splits the data into k clusters. It chooses k centroids that minimise the within-cluster sum of squared errors, $WSSSE = \sum_{j=1}^{k} \sum_{x \in C_j} \| x - \mu_j \|^2$, by repeating two steps (Lloyd's algorithm):

1. **Assign**: put each point in the cluster of its nearest centroid (Euclidean distance, from Q1).
2. **Update**: move each centroid to the mean of its assigned points.

These two steps repeat until the assignments stop changing. Spark initialises the centroids with **k-means||**, a parallel version of k-means++, and each iteration is one map (assign) plus one reduce (sum the points per cluster). This makes K-Means a natural fit for MapReduce and Spark.

Hand computation (one iteration, k = 2). The four flowers are iris rows 1 and 44 (setosa) and rows 51 and 78 (versicolor), using petal length and width. The initial centroids are C1 = flower 1 and C2 = flower 51.

Flower (petal L, W)	d(C1)	d(C2)	Assigned
1 (1.4, 0.2)	0	$\sqrt{(3.3^2+1.2^2)} = 3.5114$	C1
44 (1.6, 0.6)	$\sqrt{(0.2^2+0.4^2)} = 0.4472$	$\sqrt{(3.1^2+0.8^2)} = 3.2016$	C1
51 (4.7, 1.4)	3.5114	0	C2
78 (5.0, 1.7)	$\sqrt{(3.6^2+1.5^2)} = 3.9000$	$\sqrt{(0.3^2+0.3^2)} = 0.4243$	C2

The new centroids are C1 = mean(flower 1, flower 44) = **(1.5, 0.4)** and C2 = mean(flower 51, flower 78) = **(4.85, 1.55)**. Recomputing the distances does not change any assignment, so the algorithm has converged. The script prints these exact numbers.

Choosing k (elbow method + silhouette), computed with Spark on all 150 flowers:

k	WSSSE	Silhouette
2	152.348	**0.8504**
3	**78.851**	0.7357
4	57.256	0.6723
5	52.945	0.6158
6	39.354	0.5479

WSSSE roughly halves from k = 2 to k = 3 and then decreases much more slowly: the “elbow” is at k = 3. The silhouette is highest at k = 2, because *versicolor* and *virginica* overlap and form one well-separated blob next to *setosa*. The two criteria disagree, which is common on real data. Domain knowledge (three species) settles it, so **k = 3**.

Spark result (`KMeans(k=3, seed=1)`, converged in 3 iterations):

Cluster	Centroid (sepal L, sepal W, petal L, petal W)	setosa	versicolor	virginica
0	(5.902, 2.748, 4.394, 1.434)	0	**48**	14
1	(5.006, 3.428, 1.462, 0.246)	**50**	0	0
2	(6.850, 3.074, 5.742, 2.071)	0	2	**36**

WSSSE = 78.851 and silhouette = 0.7357. The **purity** is 134/150 = **0.8933**: setosa is found perfectly, and the 16 errors are versicolor/virginica flowers in the region where the two species overlap.

Strengths and weaknesses. K-Means is simple, fast ($O(nkt)$) and scales well. However, you must choose k in advance, and it assumes spherical clusters of similar size. It is also sensitive to outliers and to feature scale, and it gives only hard assignments.

3.4 Algorithm 2 — Bisecting K-Means (hierarchical, divisive)

Idea. Bisecting K-Means is a **top-down hierarchical** method:

1. Start with all points in one cluster.
2. Pick a cluster to split (Spark splits divisible clusters level by level, preferring the larger ones).
3. Split it into two with ordinary 2-means.
4. Repeat until there are k leaf clusters or no cluster can be divided.

The result is a binary **tree (dendrogram)** of clusters, so you can cut it at different levels. Each split only runs 2-means on a subset of the data, so it is often faster than K-Means for large k. It is also less dependent on random initialisation.

How the hierarchy unfolds on Iris (centre = sepal L, sepal W, petal L, petal W):

k	Clusters (centre, size)
2	(5.01, 3.37, 1.56, 0.29) n=53 · (6.30, 2.89, 4.96, 1.70) n=97
3	(5.01, 3.37, 1.56, 0.29) n=53 · (5.95, 2.77, 4.45, 1.45) n=59 · (6.85, 3.07, 5.74, 2.07) n=38

The first cut separates the small-petal flowers (setosa plus 3 others) from all the rest. The second cut splits the large group of 97 into a medium-petal and a large-petal cluster. Each cut is an ordinary 2-means like the hand computation in 3.3.

Spark result (`BisectingKMeans(k=3, seed=1)`):

Cluster	setosa	versicolor	virginica
0	**50**	3	0
1	0	**45**	14
2	0	2	**36**

SSE = 84.204, silhouette = 0.7232 and purity = 131/150 = **0.8733**. This is slightly worse than K-Means (SSE 78.851). The first split put 3 versicolor flowers in the setosa branch, and **bisecting never undoes an earlier split**. That greedy behaviour is the method's main weakness.

Strengths and weaknesses. Bisecting K-Means gives a hierarchy and needs fewer distance computations per step. Bad early splits are never corrected. Classic *agglomerative* (bottom-up) hierarchical clustering needs $O(n^2)$ memory and does not scale, which is why Spark offers the divisive version.

3.5 Algorithm 3 — Gaussian Mixture Model (probabilistic, soft clustering)

Idea. A GMM assumes the data was generated by k Gaussian distributions with mixing weights π_j, means μ_j and covariance matrices Σ_j:

$$p(x) = \sum_{j=1}^{k} \pi_j \, \mathcal{N}(x \mid \mu_j, \Sigma_j)$$

The parameters are estimated by **Expectation-Maximisation (EM)**:

- **E-step**: compute each point's *responsibility*, the probability that it belongs to each component: $\gamma_{ij} = \frac{\pi_j \mathcal{N}(x_i \mid \mu_j, \Sigma_j)}{\sum_l \pi_l \mathcal{N}(x_i \mid \mu_l, \Sigma_l)}$.
- **M-step**: re-estimate π_j, μ_j and Σ_j as responsibility-weighted averages.
- Repeat until the log-likelihood stops improving.

Unlike K-Means, every point gets a **probability of membership** in every cluster (soft clustering). The full covariance matrices also allow **elliptical** clusters of different sizes and orientations. K-Means is the special case of a GMM with equal spherical covariances and hard assignments.

Initialisation matters. EM only reaches a *local* optimum, so the script fits 10 random initialisations:

Seeds	Log-likelihood	Cluster sizes	Note
3, 4, 7	−180.19	50 / 55 / 45	the solution kept
10	−189.50	53 / 47 / 50	
1, 9	−192.65	24 / 50 / 76	
2	−193.21	29 / 50 / 71	
5, 8	−194.64	13 / 50 / 87	rejected: a component holds < 10%
6	**−133.69**	49 / **3** / 98	rejected: a component holds < 10%

Seed 6 has by far the *highest* likelihood but is a **degenerate** solution. One Gaussian collapsed onto 3 flowers, and its nearly singular covariance makes the density, and so the likelihood, very large. This is a known failure mode of EM for GMMs. The script keeps the highest-likelihood fit in which every component holds at least 10% of the data.

Spark result (`GaussianMixture(k=3, seed=7)`):

Component	Weight	Mean (sepal L, sepal W, petal L, petal W)	setosa	versicolor	virginica
0	0.333	(5.006, 3.428, 1.462, 0.246)	**50**	0	0
1	0.368	(6.545, 2.949, 5.480, 1.985)	0	5	**50**
2	0.299	(5.915, 2.778, 4.202, 1.297)	0	**45**	0

The log-likelihood is −180.19 and the silhouette is 0.6503. The purity is 145/150 = **0.9667**, the best of the three algorithms. Its full covariances fit the elongated, overlapping versicolor and virginica groups better than K-Means' spheres. The silhouette score is lower because silhouette itself assumes compact, round clusters.

Soft memberships. Nine flowers have a maximum membership probability below 0.95, and all of them lie in the versicolor/virginica overlap. For example:

Flower	Species	Petal L, W	P(comp 0)	P(comp 1)	P(comp 2)
78	versicolor	5.0, 1.7	0.000	**0.671**	0.329
134	virginica	5.1, 1.5	0.000	**0.784**	0.216
85	versicolor	4.5, 1.5	0.000	0.153	**0.847**

K-Means would give each of these one hard label. The GMM reports that the assignment is uncertain.

Old Faithful: a hand-checkable E-step. The script fits a 2-component GMM to the 272 eruption durations:

Component	Mean (min)	Variance	Weight
0 ("short")	2.0186	0.0555	0.3484
1 ("long")	4.2733	0.1910	0.6516

For an eruption lasting $x = 2.8$ minutes:

- $\pi_0 \, \mathcal{N}(2.8 \mid 2.0186, 0.0555) = 0.3484 \times 0.006927 = 0.002414$
- $\pi_1 \, \mathcal{N}(2.8 \mid 4.2733, 0.1910) = 0.6516 \times 0.003110 = 0.002026$
- $\gamma_0 = 0.002414/(0.002414 + 0.002026) = 0.5436$, which matches Spark.

Eruption (min)	1.8	2.8	3.0	3.2	4.5
P(short)	1.0000	**0.5436**	0.0117	0.0001	0.0000
P(long)	0.0000	**0.4564**	0.9883	0.9999	1.0000

A 2.8-minute eruption is close to a coin flip between the two eruption types. Only 12 of the 272 eruptions fall between 2.5 and 3.5 minutes.

3.6 Comparison of the three algorithms

	K-Means	Bisecting K-Means	Gaussian Mixture
Type	Partitional, centroid-based	Hierarchical (divisive)	Model-based, probabilistic
Assignment	Hard	Hard	**Soft** (probabilities)
Cluster shape	Spherical, similar size	Spherical	Elliptical, any size and orientation
Objective	Minimise WSSSE	Repeated 2-means splits	Maximise likelihood (EM)
Output	k centroids	Tree of clusters + centroids	Weights, means, covariances
Cost	Low	Low to medium	Higher (covariance per component)
Sensitivity	Initialisation, outliers, scale	Early greedy splits	Initialisation; can collapse to a few points
Iris (k=3): SSE / WSSSE	78.85	84.20	—
Iris (k=3): silhouette	0.7357	0.7232	0.6503
Iris (k=3): purity vs species	0.8933	0.8733	**0.9667**

Question 4 — Classification

4.1 What is classification?

Classification is a *supervised* learning task. It learns, from labelled training examples (x_i, y_i), a function that assigns a new observation x to one of a fixed set of **discrete classes** $y \in \{c_1, \ldots, c_m\}$. Examples:

- Is an email spam or ham?
- Is a transaction fraudulent?
- Will it rain tomorrow?
- Which digit (0–9) is in this image?

The workflow is:

1. Collect labelled data.
2. Engineer the features (`VectorAssembler`, `StringIndexer`, `OneHotEncoder`, scaling).
3. Split the data into training and test sets.
4. **Train** the model.
5. **Predict** on the test set.
6. **Evaluate** with a confusion matrix, accuracy, precision, recall, F1 and ROC-AUC.

A *binary* problem has two classes and a *multiclass* problem has more. Regression is different: it predicts a *continuous* value.

4.2 The three algorithms in words

Naïve Bayes is a **probabilistic, generative** classifier based on Bayes' theorem:

$$P(c \mid x) = \frac{P(c)\,P(x \mid c)}{P(x)} \;\propto\; P(c)\prod_j P(x_j \mid c)$$

It is "naïve" because it assumes the features are conditionally independent given the class, so the joint likelihood becomes a product of one-feature terms. Training is just counting (for categorical or multinomial data) or estimating a mean and variance per class and feature (Gaussian NB). This takes a single pass over the data, which is extremely fast and scalable. **Laplace smoothing** adds 1 to each count so that an unseen value does not make the whole product zero.

Support Vector Machine (SVM) is a **geometric, discriminative** classifier. It finds the **hyperplane** $w \cdot x + b = 0$ that separates the classes with the **largest margin** $2/\|w\|$. Only the points that lie on the margin, the **support vectors**, determine the boundary. The **soft-margin** version allows some violations through the hinge loss $\max(0, 1 - y(w \cdot x + b))$ plus a regularisation term $\lambda \|w\|^2$. The **kernel trick** (polynomial, RBF) handles non-linear boundaries. Spark MLlib provides a distributed **linear** SVM (`LinearSVC`) trained with OWL-QN.

Random Forest is an **ensemble of decision trees**, an example of bagging:

- Each tree is trained on a **bootstrap sample** (rows drawn with replacement).
- At each split, a tree considers only a **random subset of features**.
- Each split is chosen to reduce impurity the most, measured by Gini $= 1 - \sum_c p_c^2$ or by entropy.
- The forest predicts by **majority vote** of the trees, or by averaging their class probabilities.

The two kinds of randomness make the trees different from one another, so their errors partly cancel out. This reduces variance and overfitting compared with a single tree, and the forest also yields **feature importances**. Spark trains all the trees in parallel.

4.3 Worked example A — "Play Tennis" (hand computation + Spark)

The training data is the classic 14-day weather table of Quinlan (1986), as printed in Mitchell (1997, Table 3.2). It records outlook, temperature and humidity, and the label *play* (9 Yes, 5 No); the table's Wind column is omitted here. The task is to classify a new day: **Outlook = Sunny, Temp = Cool, Humidity = High**.

Naïve Bayes by hand

	Yes	No
Prior P(c)	9/14	5/14
P(Sunny | c)	2/9	3/5
P(Cool | c)	3/9	1/5
P(High | c)	3/9	4/5
Product	9/14 · 2/9 · 3/9 · 3/9 = **0.01587**	5/14 · 3/5 · 1/5 · 4/5 = **0.03429**
Posterior (normalised)	0.3165	**0.6835**

The prediction is **No (don't play)**. Spark computes every conditional count with DataFrame filters and gets exactly these numbers. Spark's `NaiveBayes` (multinomial, one-hot features, Laplace smoothing = 1) gives P(No) = 0.6379, which is also **No**. The probability is slightly different because smoothing pulls the estimates towards uniform.

Random Forest building block: Gini split selection, by hand and on Spark

The root node has 9 Yes and 5 No, so $Gini = 1 - (9/14)^2 - (5/14)^2 = 0.4592$.

Candidate split	Child nodes (Yes/No)	Weighted Gini	Gain
Outlook	Sunny (2/3): 0.48, Overcast (4/0): 0, Rain (3/2): 0.48	**0.3429**	**0.1163**
Humidity	High (3/4): 0.490, Normal (6/1): 0.245	0.3673	0.0918
Temp	Hot (2/2): 0.5, Mild (4/2): 0.444, Cool (3/1): 0.375	0.4405	0.0187

A single tree would split on **Outlook** first. In a Random Forest, each tree only sees a bootstrap sample and a random subset of features, so different trees split differently. A 7-tree `RandomForestClassifier` on Spark voted **[No, No, Yes, Yes, No, No, No]**: 5 votes against 2 for **No**. The forest's probability is P(No) = 0.701, which agrees with Naïve Bayes.

SVM on real data: Iris setosa vs versicolor

The data is the 100 Iris flowers of the *setosa* (label 0) and *versicolor* (label 1) species, described by petal length and petal width. These two species do not overlap:

Species	Petal length (cm)	Petal width (cm)
setosa	1.0 – 1.9	0.1 – 0.6
versicolor	3.0 – 5.1	1.0 – 1.8

So a straight line can separate them, which is the setting where the SVM's maximum-margin idea is easiest to see.

Spark `LinearSVC(regParam=0.001, standardization=False)` learned the hyperplane:

$$0.8125 \cdot \text{petal_len} + 1.7249 \cdot \text{petal_wid} - 3.3350 = 0$$

with margin width $2/\|w\| = 2/\sqrt{0.8125^2 + 1.7249^2} = 2/1.9067 = 1.0489$ cm.

Hand check of the decision function $f(x) = w \cdot x + b$:

- Flower 1 (setosa, petal 1.4, 0.2): 0.8125(1.4) + 1.7249(0.2) − 3.3350 = 1.1375 + 0.3450 − 3.3350 = −1.8525 < 0, so **setosa** ✓
- Flower 51 (versicolor, petal 4.7, 1.4):
  0.8125(4.7) + 1.7249(1.4) − 3.3350 = 3.8188 + 2.4149 − 3.3350 = +2.8988 > 0, so **versicolor** ✓

Support vectors. The flowers closest to the boundary are flower 44 (setosa, 1.6, 0.6) with $f(x) = -1.0000$ and flower 99 (versicolor, 3.0, 1.1) with $f(x) = +1.0000$. They lie exactly on the two margin lines $w \cdot x + b = \pm 1$, so they are the support vectors that fix the boundary. Moving any flower farther from the boundary would not change the model. Training accuracy is 100%.

Spark's `LinearSVC` minimises the regularised hinge loss $\frac{1}{n}\sum \max(0, 1 - y_i(w \cdot x_i + b)) + \frac{\lambda}{2}\|w\|^2$ with the OWL-QN quasi-Newton method. This is the soft-margin SVM. With a small λ it approaches the hard-margin solution, but it does not solve the exact quadratic program that a textbook SVM uses.

4.4 Worked example B — real spam e-mail, all three classifiers on Spark

Data. `spam7` contains **4,601 real e-mails** from the UCI *Spambase* collection, gathered at Hewlett-Packard Labs (Hopkins, Reeber, Forman & Suermondt, 1999). This copy, from the R package `DAAG`, keeps six of the original 57 features:

Feature	Meaning
`crl_tot`	total length of all runs of CAPITAL letters
`dollar`	% of characters that are "$"
`bang`	% of characters that are "!"
`money`	% of words that are "money"
`n000`	% of words that are "000"
`make`	% of words that are "make"

There are 2,788 ham and 1,813 spam e-mails (39.4% spam). The class means show the signal:

class	crl_tot	dollar	bang	money	n000	make
ham (0)	161.5	0.012	0.110	0.017	0.007	0.073
spam (1)	470.6	0.174	0.514	0.213	0.247	0.152

The data is split 70/30 (`randomSplit(seed=42)`) into 3,288 training and 1,313 test e-mails. All models are Spark ML `Pipelines`:

```
NaiveBayes(modelType="gaussian")                                    # raw features
NaiveBayes(modelType="gaussian") on log(1 + x) features             # see below
Pipeline([StandardScaler(withMean=True, withStd=True), LinearSVC(regParam=0.01)])
RandomForestClassifier(numTrees=100, maxDepth=8, seed=42)
```

Test-set results (from `outputs/q4_classification.txt`):

Model	Accuracy	F1	Precision (spam)	Recall (spam)	ROC-AUC
Naïve Bayes (Gaussian, raw features)	0.7685	0.7482	**0.9170**	0.4748	0.8564
Naïve Bayes (Gaussian, log(1+x) features)	0.8180	0.8104	0.8957	0.6262	0.8947
Linear SVM	0.8363	0.8307	0.9040	0.6692	0.9058
Random Forest (100 trees)	**0.8842**	**0.8832**	0.8948	**0.8112**	**0.9248**

ROC-AUC is computed from each model's spam *score*: the class-1 probability for NB and RF, and the signed distance $w \cdot x + b$ for the SVM, which has no probability output.

Confusion matrices (rows = actual, columns = predicted), 778 ham and 535 spam in the test set:

	NB raw: ham	NB raw: spam	NB log: ham	NB log: spam	SVM: ham	SVM: spam	RF: ham	RF: spam
actual ham	755	23	739	39	740	38	727	51
actual spam	281	254	200	335	177	358	101	434

What each fitted model learned:

- **Naïve Bayes** stores class priors (0.611 ham, 0.389 spam) and, for every class and feature, a mean and variance. For example, the mean `crl_tot` is 163.9 for ham and 467.1 for spam. Gaussian NB assumes each feature is *normally* distributed within a class, but these counts and percentages are mostly 0 with a long right tail. The badly fitting Gaussians make NB cautious: it misses 281 of 535 spams (recall 0.47). Taking $\log(1 + x)$ makes the features less skewed and raises accuracy from 0.769 to 0.818. **Checking the model's assumptions matters.**
- **SVM** learned one hyperplane on the standardised features, with weights dollar 1.181, money 0.990, bang 0.935, n000 0.820, crl_tot 0.007, make −0.022 and $b = 0.098$. A linear boundary cannot capture thresholds such as "any `$` at all is suspicious", which limits its recall.
- **Random Forest** fits non-linear, axis-aligned rules and handles the skewed features without any transformation. Its feature importances are bang 0.375, dollar 0.318, crl_tot 0.123, money 0.116, n000 0.052, make 0.016: "!" and "$" are the strongest spam signals. It has the best accuracy, recall and AUC, and its errors are the most balanced (51 false alarms, 101 missed spams).

Where the models disagree. The three classifiers (NB on raw features, SVM and RF) disagree on 270 of the 1,313 test e-mails. The first three that the script prints are spams with weak signals, such as `crl_tot = 112, dollar = 0.054, bang = 0.164`. NB and SVM call them ham and only the forest catches them. A tree can learn that small but non-zero values of `$` and `!` already indicate spam.

Interpretation. On this real data, the features are skewed, the class boundary is non-linear, and the features interact, so the ensemble wins. NB has the highest precision (few false alarms) and is the cheapest to train, which matters for a spam filter that must not lose real mail. The SVM sits in between.

4.5 Distinguishing Naïve Bayes, SVM and Random Forest

Aspect	Naïve Bayes	SVM	Random Forest
Family	Probabilistic, **generative** (models $P(x \mid c)$)	Geometric, **discriminative** (models the boundary)	**Ensemble** of decision trees (bagging)
Core principle	Bayes' theorem + conditional independence	Maximum-margin separating hyperplane	Majority vote of de-correlated trees
Decision boundary	Linear (multinomial) or quadratic (Gaussian)	Linear; non-linear with kernels	Piecewise axis-aligned, highly non-linear
Training	One pass of counting (very fast)	Convex optimisation (iterative)	Grow many trees in parallel (heavier)
Key hyperparameters	Smoothing, model type	C / regParam, kernel	numTrees, maxDepth, featureSubsetStrategy
Output	Calibrated-ish class probabilities	Signed distance / score (no native probability)	Vote fractions as probabilities + feature importance
Feature scaling needed?	No	**Yes** (distance-based)	No
Handles feature interactions	No (independence assumption)	Only via kernels	**Yes**, naturally
Robustness	Works with little data; hurt by correlated features	Robust in high dimensions; sensitive to C and scaling	Robust to outliers and noise; resists overfitting
Interpretability	High (inspect probabilities)	Medium (weights for linear)	Medium (feature importances, but many trees)
Best for	Text and spam, real-time, very high-dimensional sparse data	High-dimensional data with a clear margin (text, bioinformatics)	Tabular data with mixed types and non-linear effects
Spark MLlib class	`NaiveBayes` (multinomial, Bernoulli, Gaussian, complement)	`LinearSVC` (binary, linear)	`RandomForestClassifier`
Result on Play Tennis query	No (P = 0.684)	—	No (5 of 7 trees)
Spam (spam7) test accuracy / AUC	0.769 / 0.856 (0.818 / 0.895 with log features)	0.836 / 0.906	**0.884 / 0.925**

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